BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (31/08/2026 – 06/09/2026)

WEEK 6

PROJECT NAME	CHATBOT FOR STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE QUERY

PROJECT DESCRIPTION :	DESIGN AND IMPLEMENT GRAPHICAL USER INTERFACE FOR AN OPEN SOURCE FIREWALL
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31/08/2026	• Continued working on the architecture of the knowledge-based chatbot. • Explored approaches for handling queries across structured and unstructured data.
01/09/2026	• Worked on understanding how structured data can relate to the existing retrieval pipeline. • Explored different methods for converting user queries into appropriate database queries.
02/08/2026	• Started exploring knowledge graph integration for retrieving relationships between different entities. • Reviewed different tools and frameworks for implementing the knowledge graph component.
03/08/2026	• Worked on the initial integration approach for structured data, unstructured documents, and knowledge graphs. • Tested different query types to understand how the system should route queries to the relevant data source.
04/08/2026	**Holiday (Krishna Janmashtami)**
05/08/2026	• Continued refining the overall chatbot architecture and query processing workflow.
06/08/2026	**Holiday (Sunday)**

Next Week Plan	• Continue implementing the query processing pipeline

REFERENCE:

https://microsoft.github.io/graphrag/get_started/

https://neo4j.com/blog/developer/rag-tutorial/

https://microsoft.github.io/graphrag/index/default_dataflow/

https://huggingface.co/sentence-transformers

https://www.youtube.com/watch?v=QojA5urIDzA

https://www.youtube.com/watch?v=JfSmffOyV-8

SCREEN SHOTS:

```python
def chunk_text(text, chunk_size=CHUNK_SIZE, overlap=CHUNK_OVERLAP):
    """Split text into overlapping chunks, breaking at sentence ends
    where possible so chunks stay readable."""
    text = re.sub(r"\s+", " ", text.strip())
    if len(text) <= chunk_size:
        return [text]

    chunks = []
    start = 0
    while start < len(text):
        end = start + chunk_size
        if end < len(text):
            sentence_end = text.rfind(".", start, end)
            if sentence_end > start + chunk_size // 2:
                end = sentence_end + 1
        chunk = text[start:end].strip()
        if chunk:
            chunks.append(chunk)
        start = end - overlap
        if start >= len(text):
            break
    return chunks
```

```
retrieval_eval.py
Proper retrieval-quality evaluation using standard IR metrics, against
the hand-written gold_dataset.py (not just sample smoke-test questions).

Metrics computed:
  - Recall@k   : did the correct document appear ANYWHERE in the top-k
                 results? (out of all correct answers, how many did we find)
  - Precision@k: of the top-k results returned, what fraction were correct?
  - MRR        : Mean Reciprocal Rank - how high was the FIRST correct
                 result, averaged across all questions (1.0 = always rank 1)

For out-of-scope questions (expected_source=None), "correct" instead means
the top result's similarity stays below CONFIDENCE_THRESHOLD - i.e. the
system correctly declines to confidently answer.
```